Project
Data Analytics Portfolio

E-commerce Customer Purchase Behavior Analysis

An exploratory analysis of customer purchasing behavior, sales performance, product performance, and geographic markets using e-commerce transaction data

Tools
Python · Pandas · Matplotlib · Seaborn · Jupyter Notebook · Microsoft Excel (Pivot Tables)

by :
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I. Executive Summary

This project analyzes e-commerce transaction data to understand sales performance, customer purchasing behavior, product performance, and geographic market characteristics.

After data cleaning and validation, the final dataset contains 524,558 transaction records representing 19,865 unique transactions, with 4,335 identified customers, 3,918 unique StockCodes, 4,023 product descriptions, and 38 countries.

The business generated approximately £10.54 million in revenue during the analyzed period. Revenue and transaction volume increased substantially toward the end of 2011, reaching their highest levels in November 2011, with approximately £1.50 million in revenue and 2,761 transactions.

Customer revenue is relatively distributed, with the top 10 customers contributing 14.56% of total revenue. At the product level, REGENCY CAKESTAND 3 TIER generated the highest revenue among physical products at £174,156.54, while PAPER CRAFT, LITTLE BIRDIE recorded the highest quantity sold at 80,995 units.

Geographically, the United Kingdom is the largest market by revenue, transaction volume, and identified customer base. EIRE recorded the highest average spending per identified customer at £87,296.04, although this figure is based on only three identified customers and should therefore be interpreted cautiously.

II. Project Overview & Business Questions

2.1 Project Objective

The objective of this project is to analyze customer purchase behavior and sales performance in an e-commerce business, identify important patterns across customers, products, and geographic markets, and translate the findings into actionable business recommendations.

2.2 Business Questions

Sales Performance

1. How does revenue change over time?
2. How does transaction volume change over time?
3. What is the typical order value?

Customer

4. How concentrated is revenue among customers?
5. Which customers purchase most frequently?

Product

6. Which products generate the highest revenue and sales quantity?
7. Do high volume products also generate high revenue?

Geographic

8. Which countries contribute the most revenue, transactions, and customers?
9. Which countries have the highest average customer spending?

III. Dataset & Data Preparation

3.1 Dataset Overview

Metric	**Final Value**
Transaction records	524,558
Unique transactions	19,865
Identified customers	4,335
Unique StockCodes	3,918
Product descriptions	4,023
Countries	38
Variables	9

The dataset covers transactions from 1 December 2010 to 9 December 2011

3.2 Data Cleaning

Cleaning Step	**Records**
Original dataset	541,909
Missing Description	1,454
Unit Price = 0	2,515
Cancelled records	9,288
Duplicate records	5,226
Operational records removed	320
Final cleaned records	**524,558**

After cleaning, no cancelled transactions, non positive prices, non positive quantities, missing description, or duplicate records remained. CustomerID missing values were retained because these records still contain useful transaction level information.

IV. Sales Performance Analysis

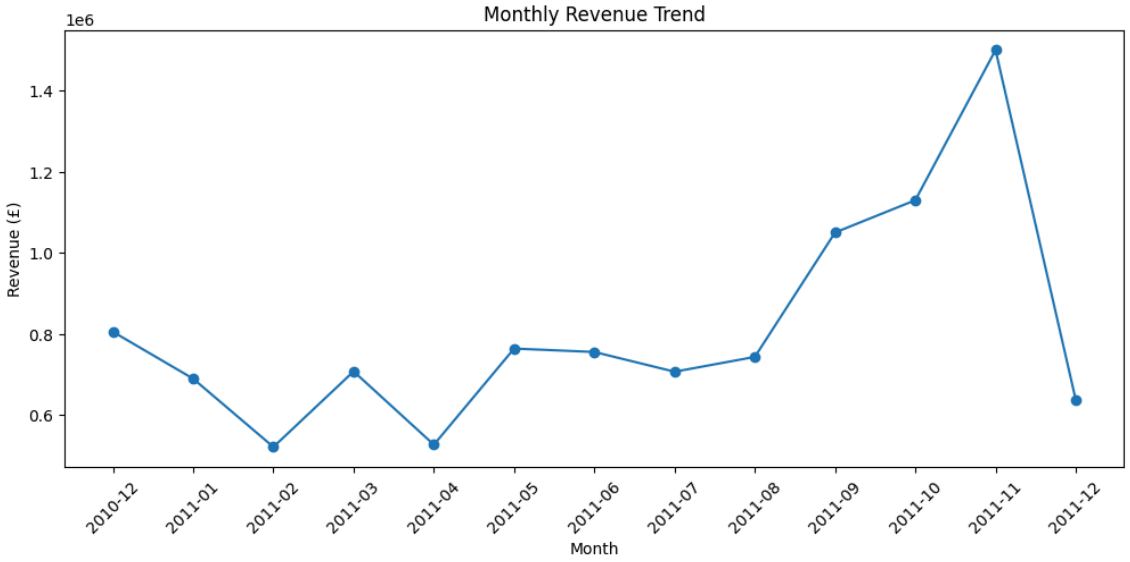
4.1 Overall Revenue

KPI :

Total Revenue: £10.54 million

4.2 Monthly Revenue

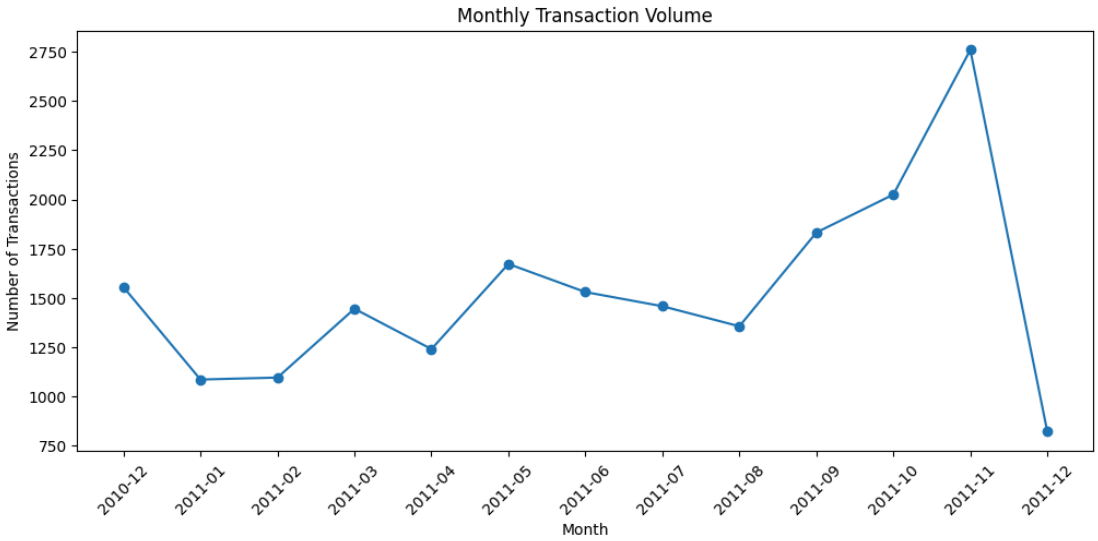
Fig 1. Monthly Revenue Trend



Revenue increased substantially toward the end of 2011, reaching its highest level in November 2011 at £1,500,554.36. The lowest monthly revenue was recorded in February 2011 at £521,308.11

4.3 Transaction Volume

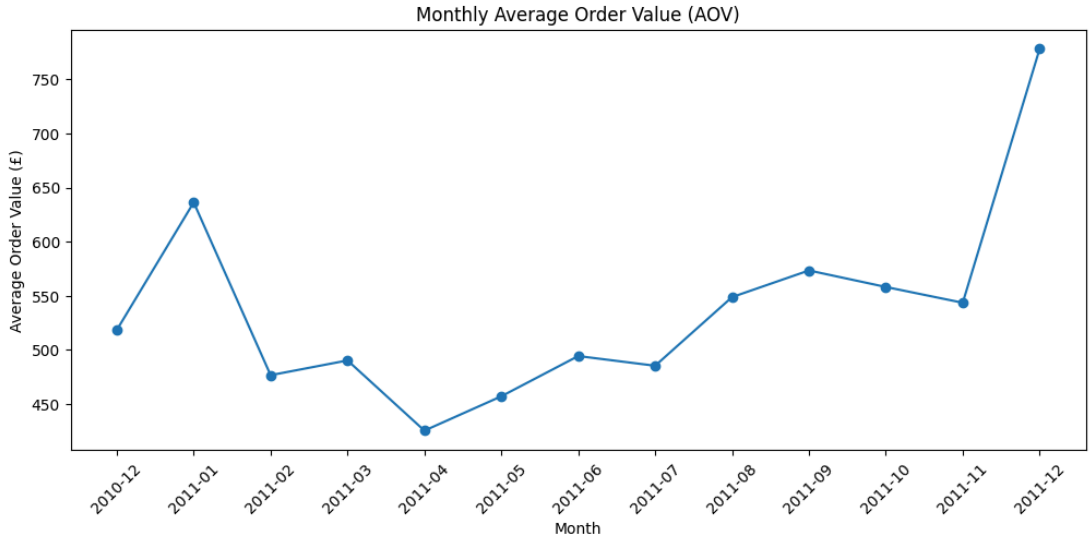
Fig 2. Monthly Transaction Volume



November 2011 > 2,761 transactions
February 2011 > 1,094 transactions

4.4 Average Order Value

Fig 3. Monthly AOV



Metric	**Month**	**Value**
Highest AOV	December 2011	£ 778.21

Lowest AOV	April 2011	£ 425.23

Revenue and transaction volume both increased substantially toward the end of 2011, reaching their highest levels in November. Since AOV did not increase consistently over the same period, the increase in revenue appears to have been primarily driven by higher transaction volume rather than substantially larger orders.

V. Customer Analysis

5.1 Customer Overview

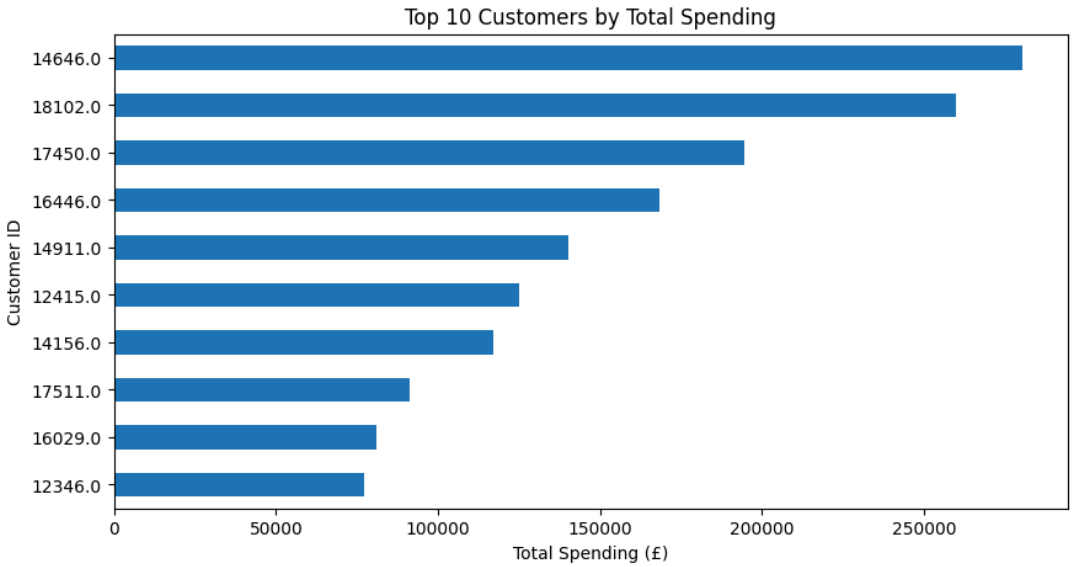
4,335 identified customers

Average spending:
£ 2,037.78

Top 10 customer contribution:
14.56% of total revenue

5.2 Top Customers

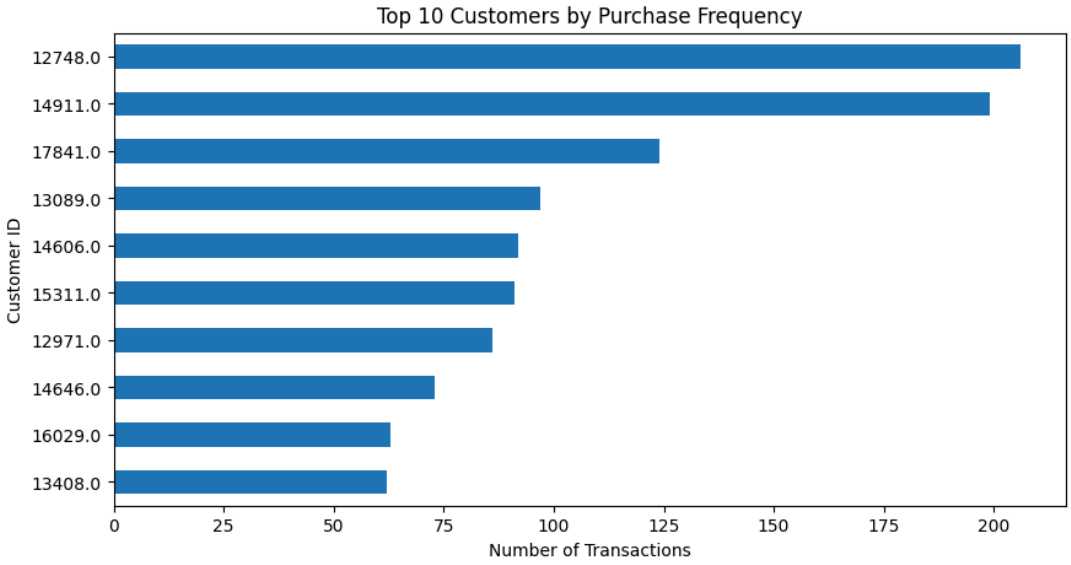
Fig 4. Top 10 Customers by Spending



The top 10 customers contribute only 14.56% of total revenue, indicating relatively low customer revenue concentration.

5.3 Purchase Frequency

Fig 5. Top 10 Customers by Purchase Frequency



Customer 12748 has the highest purchase frequency with 206 transactions. However, purchase frequency does not necessarily correspond to the highest total spending, indicating that customer loyalty and monetary value should be evaluated separately.

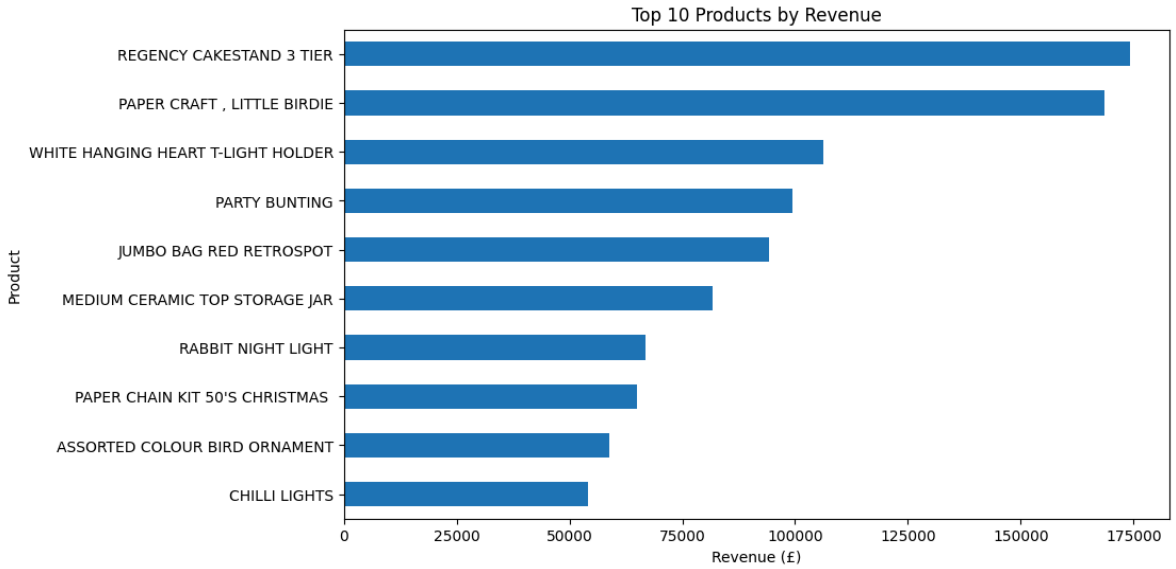
VI. Product Analysis

6.1 Product Definition

For product performance analysis, shipping and service related items such as POSTAGE and DOTCOM POSTAGE are excluded from the physical product ranking. This prevents service charges from being interpreted as merchandise performance.

6.2 Product by Revenue

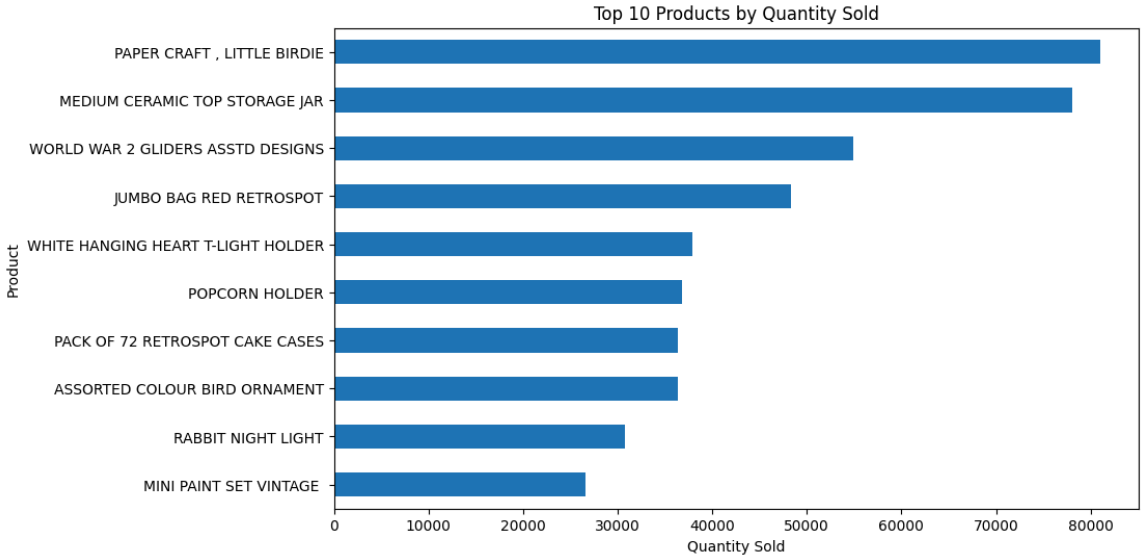
Fig 6. Top 10 Product by Revenue



REGENCY CAKESTAND 3 TIER generated the highest revenue among physical products, contributing approximately £ 174,156.54.

6.3 Products by Quantity

Fig 7. Top 10 Product by Quantity



#1 PAPER CRAFT, LITTLE BIRDIE > 80,995 units

6.4 Revenue vs Quantity

Six out of the top 10 products by quantity also appear among the top 10 revenue generating products, representing a 60% overlap. This indicates that several high demand products also contribute strongly to revenue, although sales volume alone does not determine revenue performance.

VII. Geographic Analysis

7.1 Market Overview

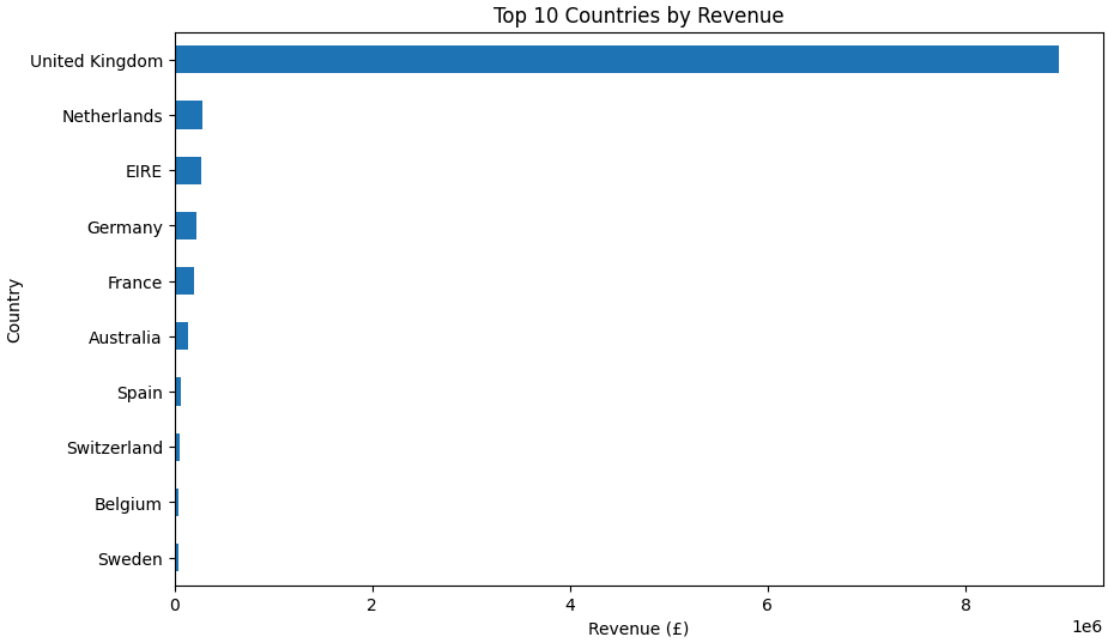
38 countries

United Kingdom

Metric	**Result**
Revenue	**£8,941,988.24**
Transactions	**17,954**
Identified customers	**3,917**

7.2 Revenue by Country

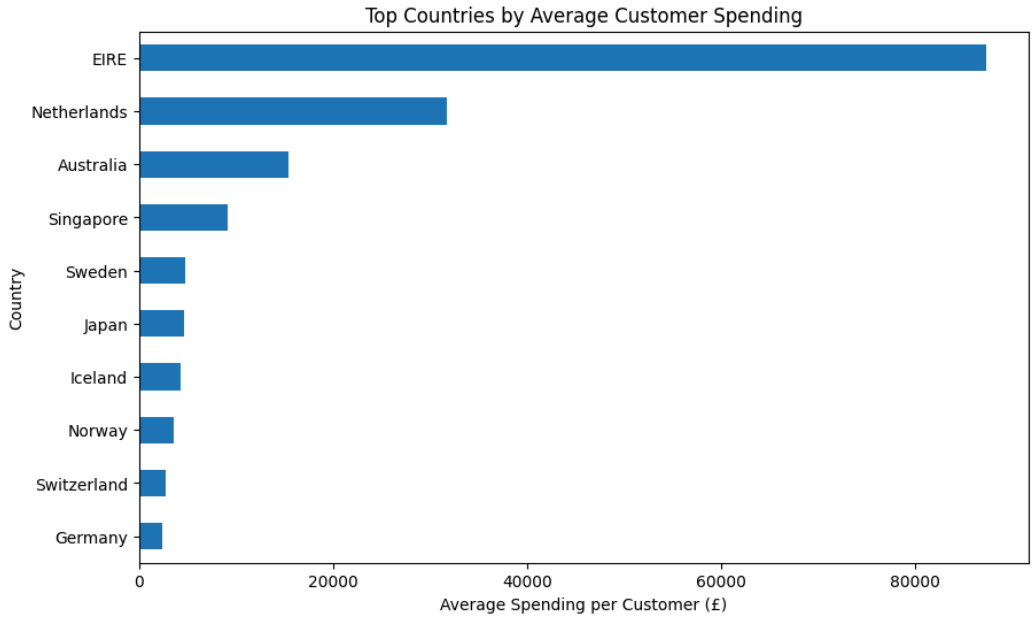
Fig 8. Top 10 Countries by Revenue



United Kingdom £8,941,988.24

7.3 Customer Spending by Country

Fig 9. Average Customer Spending by Country



EIRE recorded the highest average spending per identified customer at £87,296.04. However, this result is based on only three identified customers and should therefore be interpreted cautiously.

VIII. Key Findings

Business Scale

524,558 records across 19,865 transactions, 4,335 identified customers, 3,918 StockCodes, 4,023 product descriptions, and 38 countries.

Sales Trend

Revenue and transaction volume increased substantially toward the end of 2011, peaking in November.

Customer Behavior

Top 10 customers contribute only 14.56% of revenue, indicating relatively low customer concentration.

Customer Loyalty ≠ Customer Value

Customer 12748 made the most purchases (206 transactions), but purchase frequency does not necessarily indicate the highest monetary value.

Product Performance

60% of the top 10 quantity products also appear in the top 10 revenue list.

Geographic Market

The United Kingdom is the largest market by revenue, transaction volume, and identified customer base, while EIRE has the highest average spending per identified customer, based on only three customers.

IX. Business Recommendations

Customer Strategy

Maintain a broad customer acquisition and retention strategy rather than relying heavily on a small group of high-value customers. At the same time, frequently purchasing customers should be targeted with loyalty initiatives to encourage repeat purchases and increase their long-term customer value.

Product Strategy

Prioritize inventory availability and promotional activities for products that perform strongly in both sales quantity and revenue. Product performance should be evaluated using both metrics rather than either one alone.

Geographic Strategy

Maintain customer retention and market coverage in the UK, which represents the company's largest market by revenue and customer base. EIRE could be explored for higher-value customer opportunities, but further customer data should be collected before making major strategic decisions because the current average is based on only three identified customers.

Seasonal Strategy

Increase inventory readiness and promotional planning ahead of the September–November period to prepare for higher transaction volume. However, the observed seasonal pattern should be monitored across additional periods before being treated as a recurring trend.

Conclusion

This analysis identified several important patterns in the e-commerce business. Revenue is relatively distributed across customers, while purchasing activity increases substantially toward the end of the year. Product performance varies depending on whether it is measured by quantity or revenue, highlighting the importance of evaluating both metrics. Geographically, the UK represents the company's largest market, while smaller markets such as EIRE show potentially higher customer value but require further data before strategic conclusions can be made.

Overall, the findings provide a data-driven basis for customer retention, product prioritization, geographic strategy, and seasonal planning.

Appendix :

Link repo github : GitHub - aliciaajsp/EcommerceCustomerPurchaseAnalysis · GitHub